

Date: _____

Name :

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Registration #

FA22-BCS-086

Section :

'B'

Submitted to:

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Darsi

Date: _____

Question No 1

(1)

Database Management System:

Database management system is a software that enable user to create, maintain, manipulate and delete database.

DBMS is used to

- Define database
- Construct database
- Manipulate database

(2)

Instance	Schema
The actual data store in database in a particular moment in time Also called database state	Schema is a blueprint that defines how data is organized including relationships, constraints and data type.
" "	

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Instance is an
actual data
" "

Schema is the design
that governs how data is
stored

(3)

Disadvantages of file Processing System.

1. Data Redundancy
2. Data inconsistency
3. Data Isolation
4. Integrity Problem
5. Lack of Concurrent Access
6. Atomicity Problem
7. Security Problem
8. Difficult to access

(4)

1. Data independence :

It is the ability to change the schema at one level without having to make change at the next higher level.

When schema is changed at lower level then only the mappings between this schemas and higher level schemas.

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Hence the application programs need not to be changed since they refer to external schema.

(5)

Two tier client-server architecture

Three tier client-server architecture.

• In this architecture, the client communicate directly with the server.

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• Two tiers are

- 1) Presentation layer
- 2) Data Layer

• In this architecture, an additional middle layer exist between client and server which enhance security.

• Three tiers are :

- 1) Presentation Layer
- 2) Application Layer
- 3) Data Layer

Question No 2.

A

ERD diagram.

• A college contains many department Darsi

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- Each department offers any number of courses
- Many instructors can work in a department
- An instructor can work only in one department
- For each department there is a head
- An instructor can (work) head only one department
- Each instructor can take any no. of courses
- A course can be taken by only one instructor
- A student can enroll for any number of course
- Each course can have any no. of students

Entities are.

Department, College, Course, instructor, student.

Relationship

E ₁	Relation	E ₂
College	contain	Department
Department	offer	Course
Dep Student	enroll	course
Instructor	work in	Department
Instructor	head	Department
Instructor	take	Course.

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Entity Relationship Diagram.

